

SUN YUSHI

PERSONAL DATA

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EXPERIENCE

AUG 2025 - PRESENT Senior Researcher (Project UP), **Tencent Games**, **LightSpeed Studios**, Shenzhen
SEP 2021 - OCT 2025 PhD in Computer Science, **HKUST**, Hong Kong
Advised by Prof. Lei Chen
FEB 2025 - MAY 2025 Research Intern, **Ant Research** (Multi-KG-based RAG)
SEP 2017 - JUN 2021 BSc in Computer Science and Mathematics, **HKUST**, Hong Kong
SEP 2019 - FEB 2020 [Exchange student](#) at **ETH**, Zurich
JULY 2019 [Exchange student](#) at **NTU**, Singapore

RESEARCH VISION

My research vision is to **make AI systems reason reliably over data**. Rooted in data management, I develop retrieval-augmented generation (RAG) and agent memory that turn raw data and knowledge into trustworthy LLM and agent behavior: from question answering to scientific discovery.

Research topics: Retrieval-Augmented Generation (RAG), Agent Memory, Knowledge Graphs, Text-to-SQL, Multimodal LLMs, AI4Science, Data Management.

SELECTED PUBLICATIONS AND PREPRINTS

Notation: * equal contribution, + corresponding author.

- 2026 **Are LLMs Ready for Scientific Discovery? A Capability-Oriented Benchmark for AI Scientists**
([paper](#))
Chuhan Shi, Xiaoquan Ren, Sicheng Song, Haobo Li, Rui Sheng⁺, **Yushi Sun⁺**
Preprint 2026
[AI4S]
- 2026 **When Classic Cache Policies Fail: Learning-Augmented Replacement for Semantic Retrieval Buffers**
([paper](#))
Yushi Sun^{*}, Bowen Cao^{*}, Wai Lam
Preprint 2026
[RAG, Agent Memory]
- 2026 **STALE: Can LLM Agents Know When Their Memories Are No Longer Valid?**
([paper](#))
Hanxiang Chao^{*}, Yihan Bai^{*}, Rui Sheng, Tianle Li, **Yushi Sun⁺**
Preprint 2026
[Agent Memory]

- 2026 **GRAVITY: Architecture-Agnostic Structured Anchoring for Long-Horizon Conversational Memory**
([paper](#))
Yushi Sun*, Bowen Cao*, Dong Fang⁺, Lingfeng Su, Wai Lam
Preprint 2026
[Agent Memory]
- 2026 **LakeHopper: Cross Data Lakes Column Type Annotation through Model Adaptation**
([paper](#))
Yushi Sun, Xijia Li, Nan Tang, Quanqing Xu, Chuanhui Yang, Lei Chen
Preprint 2026
[Data Management]
- 2026 **MedKGI: Iterative Differential Diagnosis with Medical Knowledge Graphs and Information-Guided Inquiring**
([paper](#))
Qipeng Wang, Rui Sheng, Yafei Li, Huamin Qu, Yushi Sun⁺, Min Zhu⁺
Preprint 2026
[Knowledge Graphs, AI4S]
- 2026 **CacheRAG: A Semantic Caching System for Retrieval-Augmented Generation in Knowledge Graph Question Answering**
([paper](#))
Yushi Sun, Lei Chen
Preprint 2026
[RAG, Knowledge Graphs]
- 2026 **VeriLLMed: Interactive Visual Debugging of Medical Large Language Models with Knowledge Graphs**
([paper](#))
Yurui Xiang, Xingyi Mao, Rui Sheng, Zixin Chen, Zelin Zang, Yuyang Wu, Haipeng Zeng, Huamin Qu, Yushi Sun⁺, Yanna Lin⁺
IEEE VIS 2026
[Knowledge Graphs]
- 2026 **Navigating the Mirage: A Dual-Path Agentic Framework for Robust Misleading Chart Question Answering**
([paper](#))
Yanjie Zhang, Yafei Li, Rui Sheng, Zixin Chen, Yanna Lin, Huamin Qu, Lei Chen, Yushi Sun⁺
ACM MM 2026
[Multimodal LLMs]
- 2026 **APEX-SQL: Talking to the Data via Agentic Exploration for Text-to-SQL**
([paper](#), [code](#))
Bowen Cao, Weibin Liao, Yushi Sun⁺, Dong Fang⁺, Haitao Li, Wai Lam
ACM SIGKDD 2026
[Text-to-SQL, Data Management]
- 2025 **KERAG: Knowledge-Enhanced Retrieval-Augmented Generation for Advanced Question Answering**
([paper](#), [code](#))
Yushi Sun, Kai Sun, Ethan Yifan Xu, Xiao Yang, Xin Luna Dong, Nan Tang, Lei Chen
EMNLP 2025
[RAG, Knowledge Graphs]

- 2024 **CRAG - Comprehensive RAG Benchmark**
([paper](#))
Xiao Yang*, Kai Sun*, Hao Xin*, **Yushi Sun***, Nikita Bhalla, Xiangsen Chen, Sajal Choudhary, Rongze Daniel Gui, Ziran Will Jiang, Ziyu Jiang, Lingkun Kong, Brian Moran, Jiaqi Wang, Ethan Yifan Xu, An Yan, Chenyu Yang, Eting Yuan, Hanwen Zha, Nan Tang, Lei Chen, Nicolas Scheffer, Yue Liu, Nirav Shah, Rakesh Wanga, Anuj Kumar, Wen-tau Yih, Xin Luna Dong
NeurIPS 2024
[RAG]
- 2024 **Are Large Language Models a Good Replacement of Taxonomies?**
([paper](#), [code](#))
Yushi Sun, Hao Xin, Kai Sun, Ethan Yifan Xu, Xiao Yang, Xin Luna Dong, Nan Tang, Lei Chen
VLDB 2024
[Knowledge Graphs]
- 2024 **Cross-domain-aware Worker Selection with Training for Crowdsourced Annotation**
([paper](#), [code](#))
Yushi Sun, Jiachuan Wang, Peng Cheng, Libin Zheng, Lei Chen, Jian Yin
ICDE 2024
[Data Management]
- 2023 **RECA: Related Tables Enhanced Column Semantic Type Annotation Framework**
([paper](#), [code](#))
Yushi Sun, Hao Xin, Lei Chen
VLDB 2023
[Data Management]

AWARDS & HONORS

- 2025 Tencent Project Up Scholar
2023 RedBird Academic Excellence Award
2022 RedBird Academic Excellence Award
2021 HKUST Academic Achievement Medal (Top 1% in HKUST)
2021 Hong Kong PhD Fellowship Scheme (HKPFS)
2021 RedBird Academic Excellence Award

ACADEMIC RECORDS

Postgraduate GPA: 4.043/4.300
Undergraduate GPA: 4.030/4.300 (First Class Honor; Dean's List of the School of Engineering)

ACTIVITIES

Co-hosting the Meta KDD Cup: CRAG: Comprehensive RAG Benchmark ([link](#))

TEACHING EXPERIENCE

- FALL 2024 Teaching Assistant of COMP 1021 Introduction to Computer Science
FALL 2022 Teaching Assistant of COMP 2711H Honors Discrete Mathematical Tools for Computer Science
SPRING 2022 Teaching Assistant of COMP 5712 Introduction to Combinatorial Optimization